

Krishi Attri

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Research Profile

Robotics researcher admitted to the Ph.D. programme at the University of Central Florida (Rehabilitation Engineering & Assistive Device Lab), currently completing an M.S. in Mechanical Engineering at Seoul National University under a Government Science Fellowship (GSFS). My research focuses on **online 3D reconstruction and pose tracking for robotic manipulation**, specifically developing visuo-tactile SLAM systems built around explicit **3D Gaussian Splatting** representations that enable robots to reconstruct and track objects under heavy occlusion during in-hand manipulation.

I am currently developing and integrating these systems as part of a humanoid robot project at the lab, fusing multimodal perception—RGB-D vision, DIGIT tactile sensing, and hand proprioception—into a shared Gaussian object state that simultaneously supports reconstruction, pose tracking, and manipulation interfaces. In parallel, I build and maintain open-source research software released on PyPI: HiCache++, a training-free DMD velocity-cache that accelerates diffusion- and flow-based image-to-3D models; splatreg, a Gaussian-splat registration library; mathlas, a formal-math MCP toolkit for AI agents; and CERT-FLOW, a certified route planner under drifting costs — three of which (splatreg, HiCache++, CERT-FLOW) are documented in solo-author engrXiv preprints (2026). My work spans GPU-accelerated algorithm development, multimodal sensor fusion, and full-stack hardware–software integration on platforms including the UR5e and Allegro Hand.

Education

Ph.D., Mechanical Engineering — Post Master's Track	<i>Aug 2026 (Incoming)</i>
<i>University of Central Florida ORCGS Doctoral Fellow Rehabilitation Engineering & Assistive Device Lab (REAL) Advisor: Professor Hwan Choi</i>	<i>Orlando, FL, USA</i>
Master of Science in Mechanical Engineering	<i>Sept 2024 – Aug 2026 (Expected)</i>
<i>Seoul National University GSFS Scholar Soft Robotics & Bionics Laboratory</i>	<i>Seoul, South Korea</i>
Bachelor of Science in Mechanical Engineering	<i>Aug 2020 – May 2024</i>
<i>Villanova University Minor in Mechatronics Control & Dynamics  Diploma</i>	<i>Villanova, PA, USA</i>
Exchange Student — B.S. Mechanical Engineering	<i>Aug 2022 – June 2023</i>
<i>Yonsei University</i>	<i>Seoul, South Korea</i>

Research Experience

Graduate Research Student

Sept 2024 – Aug 2026

Soft Robotics & Bionics Laboratory, Seoul National University

Seoul, South Korea

Humanoid Robot Project — “Alchemist” Project (MOTIE, Republic of Korea; Phase 2)

- Contributing to **Phase 2 (2024–2026)** of the “Alchemist” Project — Korea’s national high-risk, high-reward R&D program (MOTIE; Phase 1 completed 2024) — at the Soft Robotics & Bionics Lab.
- Leading integration of visuo-tactile SLAM and dexterous in-hand manipulation modules into the Phase 2 full-scale humanoid prototype.
- Coordinating cross-disciplinary teams to hit quarterly hardware–software integration milestones.

GaussianFeels: Object-Centric Gaussian SLAM for Visuo-Tactile In-Hand Manipulation (M.S. Thesis)

Advisor: Professor Yong-Lae Park, Seoul National University

- Developed GaussianFeels: an online visuo-tactile reconstruction and 6-DoF pose-tracking system built around an explicit object-centric 3D Gaussian Splatting map for robotic in-hand manipulation under heavy occlusion — no CAD model required.
- Fuses RGB-D vision, DIGIT tactile contact geometry, and hand proprioception into a single Gaussian object state maintained online, with occlusion-aware multimodal supervision and a SLAM-loop SE(3) pose tracker; reconstruction is seeded at frame 0 by an image-to-3D generative shape prior that is progressively replaced by measured geometry.
- Evaluated on the FeelSight benchmark (8 simulation + 6 real-robot objects) across its simulation, real-robot, and occlusion splits, benchmarked against the NeuralFeels and V-HOP baselines.
- **Results:** pose-mode ADD-S of 1.13 mm (simulation, 50 FPS) and 2.42 mm (coverage-adequate real objects, 37 FPS) on a single RTX 5090; map-mode observed-surface F-score@5 mm of 0.997 (sim) / 0.950 (real); lower pose error than NeuralFeels on every simulation aggregate at $7.0\text{--}12.4\times$ the per-frame speed ($5.8\text{--}8.3\times$ on real); full-surface F-score 0.997 vs. NeuralFeels’ 0.908 on the matched object, cutting reconstruction time from ≈ 52 min to 12 s; edges V-HOP on occlusion-split ADD-S (1.45 vs. 1.46 mm) at 48.8 vs. 32 FPS — all with no supplied CAD model.

PoP-SLAM: Point Cloud Projection for SLAM  [Paper](#)

Co-authors: Seongmin Jung (SNU AI), Josselin Marchand (École des Mines), Michele Lorenzo Paolicchi (Vrije Universiteit Amsterdam)

- Co-developed PoP-SLAM: a dense visual SLAM system that introduces a projection-first rendering strategy, eliminating computationally intensive nearest-neighbor searches inherent to neural point cloud-based SLAM.
- Designed GPU-vectorised point cloud projection: transforms up to $\sim 15\,000$ neural points per frame into image space using vectorised matrix multiplication, enabling real-time SLAM at **~ 4 FPS**.
- Introduced direct occlusion detection via multi-keyframe depth masking: projects neural points to nearby keyframes and retains only points consistent with measured depth across all keyframes, avoiding volume rendering entirely.
- Achieved **best ATE RMSE of 0.75 cm** on TUM-RGBD dataset, outperforming Point-SLAM, NICE-SLAM, ESLAM, and SplaTAM; competitive 0.38 cm average on Replica dataset.
- Evaluated on synthetic (Replica) and real-world (TUM-RGBD) benchmarks; demonstrated robustness across varied indoor scenes.

Undergraduate Research Intern (Winter Internship)

January 2024

Soft Robotics & Bionics Lab, Dept. of Mechanical Engineering, Seoul National University [Letter](#) Seoul, South Korea

- Participated in a team project developing a surface electromyography (sEMG) sensor with flexible design elements for enhanced wearability and adaptability.
- Contributed to research on improving sensor stretchability through PDMS (polydimethylsiloxane) and vapour-deposited silver nanoparticles.
- Assisted in applying deep learning models (CNN-GRU, CNN-RNN, ViT) to sEMG signal processing, contributing to efforts to reduce classification error and improve gesture-recognition accuracy.
- Gained proficiency in ML-based signal processing and collaborated with a team of engineers to meet project milestones.

Robotics & Mechatronics Researcher

Aug 2023 – May 2024

Villanova University [EOD robot](#) [Drone](#)

Villanova, PA, USA

- Collaborated with a Ph.D. candidate on a dissertation titled “Autonomous Localisation and Navigation in GNSS-Denied Environments,” advancing SLAM algorithms via LiDAR–camera fusion and visual odometry for a quad-wheel robot.
- Developed a full ROS-based navigation stack for the robot, implementing path planning and obstacle-avoidance algorithms.
- Applied CNN-based feature extraction and point cloud generation from fused LiDAR–camera data to support autonomous decision-making without GPS.
- Integrated and configured Arduino and Raspberry Pi microcontrollers for robust ROS communication and real-time control.
- Implemented computer vision techniques (edge detection, landmark identification) to build real-time 3D environment maps.
- Implemented a 2D histogram localisation filter and a 1D Kalman filter tracker using probabilistic motion models for improved tracking accuracy.

Publications & Software

Theses, Papers & Preprints

- **GaussianFeels: Object-Centric Gaussian SLAM for Visuo-Tactile In-Hand Manipulation.** K. Attri. M.S. Dissertation, Seoul National University, Aug 2026. *(Thesis & code release upcoming.)*
- **Registering Gaussian Splats Without the Point-Cloud Detour: Accuracy, Representation Semantics, and a Negative Result on Hypothesis-Stage Transfer.** K. Attri. engrXiv preprint, 2026. DOI: 10.31224/7313. *(Solo author; companion software: splatreg.)* [PDF](#)
- **No Single Basis Wins: A Cross-Family Study of Diffusion Feature Forecasting and the Limits of Training-Free Basis Selection.** K. Attri. engrXiv preprint, 2026. DOI: 10.31224/7309. *(Solo author; companion software: HiCache++.)* [PDF](#)
- **CERT: Certified Route Planning under Drifting Costs — Conformal Certificates, Sense-to-Certify, and the Price of Staleness.** K. Attri. engrXiv preprint, 2026. DOI: 10.31224/7306. *(Solo author; companion software: CERT-FLOW.)* [PDF](#)
- **PoP-SLAM: Point Cloud Projection for SLAM.** S. Jung, K. Attri, J. Marchand, M. L. Paolicchi. Course research paper, Seoul National University, Dec 2024. [Paper](#)

Open-Source Software (PyPI releases)

- **splatreg** — Gaussian-splat registration & merging. [PyPI](#)
- **mathlas** — airtight-math MCP toolkit for AI agents. [PyPI](#)
- **HiCache++** — training-free diffusion/flow sampler acceleration. [PyPI](#)
- **CERT-FLOW** — certified route planning under drifting costs. [PyPI](#)

Professional Experience

Indoor Farm Robotics Intern

June 2023 – Aug 2023

Area2Farms © [SILO demo](#)

Arlington, VA, USA

- Contributed to the “Silo” automation tool for the local food system supply chain.
- Gained hands-on experience in extruded aluminium construction, pneumatics, industrial robotics, Arduino/Raspberry Pi programming, and agricultural irrigation systems.
- Collaborated with a cross-functional team to prototype automation solutions for indoor vertical farming.

Product Design Intern

June 2022 – Aug 2022

Ampere LLC (Remote) [Certificate](#)

Virtual, USA

- Contributed to 3D product modelling for upcoming consumer-technology designs, including **Shower Power** and an upcoming iPhone charging product.
- Analysed structural integrity and physics constraints to ensure product performance and durability.

Distance Education Operator

Sept 2021 – Dec 2021 & Aug 2023 – May 2024

Villanova University

Villanova, PA, USA

- Provided technical support for audio/video systems used in online course production.
- Managed live-stream recording and archiving of Distance Education materials for course content reliability.

Collections & Stewardship Technician

Jan 2022 – May 2022

Villanova University

Villanova, PA, USA

- Produced digital content for the university Digital Library using scanners, digital cameras, and archiving software.
- Managed records in cataloguing and acquisitions software; supported preservation of rare and special collections materials.
- Labelled and physically prepared new and rare library materials, and coordinated binding/repair shipments of books, periodicals, and theses.

Research Projects

GaussianFeels: Object-Centric Gaussian SLAM for Visuo-Tactile In-Hand Manipulation (M.S. Thesis)

Dec 2024 – Present

Objective: Achieve robust in-hand object reconstruction and pose tracking under heavy occlusion by fusing RGB-D vision, DIGIT tactile sensing, and hand proprioception into a shared object-centric 3D Gaussian Splatting representation.

- **Approach:** Object-centric Gaussian map updated online from RGB-D, DIGIT tactile, and proprioceptive observations; occlusion-aware multimodal supervision; SLAM-loop SE(3) pose tracking (multi-residual Levenberg–Marquardt over a Gaussian-density SDF); frame-0 image-to-3D shape prior progressively replaced by measured geometry.
- **Evaluation:** FeelSight benchmark (8 sim + 6 real objects) — simulation, real-robot, and occlusion splits; benchmarked against NeuralFeels and V-HOP. Pose-mode ADD-S 1.13 mm (sim, 50 FPS) / 2.42 mm (coverage-adequate real, 37 FPS) on one RTX 5090; map F-score@5 mm 0.997 (sim) / 0.950 (real); beats NeuralFeels on every sim pose aggregate at 7.0–12.4× per-frame speed (5.8–8.3× real); full-surface F-score 0.997 vs. 0.908, reconstruction ≈ 52 min $\rightarrow$ 12 s; edges V-HOP occlusion ADD-S (1.45 vs. 1.46 mm) at 48.8 vs. 32 FPS — no supplied CAD model.

PoP-SLAM: Point Cloud Projection for SLAM (Seongmin Jung, Krishi Attri, Josselin Marchand, Michele Lorenzo Paolicchi) [Sept 2024 – Dec 2024](#) [Paper](#)

- **Innovation:** Projection-first rendering pipeline projecting $\sim 15\,000$ learned neural points/frame on GPU via vectorised matrix multiplication — no nearest-neighbor search required.
- **Occlusion Handling:** Multi-keyframe depth masking selects only points consistent with measured depth across nearby keyframes — no volume rendering required.
- **Tracking Accuracy:** ATE RMSE 0.75 cm on TUM-RGBD (best among all evaluated methods including Point-SLAM, NICE-SLAM, ESLAM, SplatTAM); 0.38 cm average on Replica.
- **Speed:** ~ 4 FPS on NVIDIA RTX 4070 ($3\times$ faster than Point-SLAM baseline); $< 3.3\%$ overhead from point pruning.
- **Datasets:** TUM-RGBD (fr1-desk, fr1-room, fr2-xyz, fr3-office) and Replica (synthetic indoor scenes).

Personal Projects

HiCache++: Training-Free DMD Velocity-Cache for Diffusion & Flow-Based 3D Generation [2026](#) [Preprint](#) [PyPI](#)

- Designed **HiCache++**, a model-agnostic, training-free DMD / exponential (Prony) velocity-cache that skips redundant denoising compute in diffusion and flow samplers — pushing the TaylorSeer $\rightarrow$ HiCache (Hermite) feature-forecasting line to its exponential endpoint. Lossless-to-near-lossless on *every* flow-matching image-to-3D generator tested: **+0.13 / +0.24** F-score over the deployed Hermite arm at skip interval 5 / 6 on Hunyuan3D-2.1, and geometry-lossless through skip 6 on Meta SAM 3D Objects at **1.56 $\times$** .
- Ran a **cross-family study** across six image-to-3D and generative backbones — Hunyuan3D-2 mini / 2.1, TRELIS v1 / TRELIS.2-4B, Meta SAM 3D, Fast-SAM3D, DiT-XL/2 — via a **14-repository adapter family** including a ComfyUI node, a SAM 3D adapter, and a TRELIS.2 adapter. Established that the optimal forecast basis is *sampler-family-dependent* (no single basis wins both flow-matching and DiT-style diffusion), and that training-free holdout selection — exact on synthetic regime switches (120/120) — does not transfer to real DiT features.
- Surfaced a benchmark-integrity finding: a one-character sign error that left a published baseline anti-extrapolative yet invisible to every end-to-end metric. Proposed directional closed-form regression tests as a community remedy and reported all affected numbers as-released alongside corrected values.
- Released open source (Python; MIT) with a PyPI package and a solo-author engrXiv preprint (*No Single Basis Wins*, 2026; DOI: 10.31224/7309); directly accelerates the image-to-3D prior used by the GaussianFeels frame-0 reconstruction branch.

splatreg & mathlas: Open-Source Research Tools [2026](#)

- **splatreg** [GitHub](#) [Preprint](#) [PyPI](#) — a pure-PyTorch library that registers and merges native 3D Gaussian-Splatting scans into one SE(3)/Sim(3) frame via a closed-form-Jacobian Gaussian-SDF residual. Matches the GeoTransformer baseline on official 3DMatch (91.5% mean / 93.5% pooled recall) and 3DLoMatch (72.5% / 74.4%) while adding the Sim(3) scale DoF it lacks; **2.9 $\times$** more accurate in rotation than existing splat tools (5.2° vs 15.3°) and **5.1 $\times$** better merge Chamfer than naive concatenation, with real-basis spherical-harmonic rotation verified to 2.4×10^{-15} . Documented in a solo-author engrXiv preprint (2026; DOI: 10.31224/7313).
- **mathlas** [GitHub](#) [PyPI](#) — an “airtight-math” MCP toolkit (12 LLM-free, data-returning tools) giving AI agents a 3.7M-document theorem search (dual-channel R@10 0.999), PSLQ/OEIS constant & sequence identification, and Lean-kernel formal verification (full proof checking) at a 0% measured false-positive rate; its self-augmenting retrieval loop surpasses TheoremSearch on its own 110-query benchmark (59.1% vs 45.0% Hit@20); listed on the official MCP registry.

CERT-FLOW: Certified Route Planning under Drifting Costs [Preprint](#) [PyPI](#)

2026

- Designed and built a planner that emits a high-probability certificate $LB \leq OPT \leq UB$ on optimal route cost every replanning round — age-weighted non-exchangeable conformal prediction over drift-adjusted residuals, with paid sensing directed at the edges that shrink the certified gap fastest.
- Backed by seven theorems (coverage with explicit staleness degradation, a certifiability threshold, and an impossibility result); coverage 0.95–1.00 on every condition run — 17 synthetic regimes, game maps, and replayed real traffic (METR-LA, PEMS-BAY) at up to 49% drift-model violation — versus 0.02–0.59 for classical replanning.
- 2–3× lower travel-regret than freshness-, uncertainty-, or chance-driven sensing at equal budget; 3.7 ms p50 per fully certified round on one CPU core, with certificate-gated static queries in 269 ns–8.7 μ s; 223 tests and 16 scripted reproduction pipelines.
- Solo-author engrXiv preprint (*CERT*, 2026; DOI: 10.31224/7306) with an open-source MIT package on PyPI.

Previous Personal Projects [Photos](#)

Summer 2021

- **Wi-Fi Drone:** Designed and programmed a Wi-Fi-controlled drone with custom stability logic.
- **Swarm Drones:** Developed multi-drone coordination algorithms for communication and synchronisation.
- **Robotic Arm:** Built a servo-actuated robotic arm programmed for pick-and-place operations.
- **3D Printer:** Assembled a 3D printer from scratch including hardware build and slicing software configuration.

Course Projects

Capstone: Plant Lifting Device for 3D Imaging [Demo](#) [Award](#) [Photos](#)

Aug 2023 – May 2024

Sponsored by a major agrochemical company — **1st Place, Most Innovative Solution**

- Led a multidisciplinary team through the full design-to-prototype lifecycle: actuator selection, motion system (belts, pulleys, drive gears), wiring, waterproofing, and plant platform engineering.
- Addressed sponsor constraints (strict size limits), plant stability risks, and motor-damage risks by developing a stabilising platform and controlled-speed mechanism.
- Applied project management across all design phases; awarded first place for Most Innovative Solution at the university capstone showcase.

Arduino Projects — Mechatronics Coursework

Jan 2022 – May 2022

- Designed and built three Arduino-based projects integrating pressure, photo, and sound sensors with actuators, motors, and LCD displays.
- Notable project: a puzzle box that randomised solution steps via algorithm and deployed a glitter-spray penalty for incorrect sequences.

Beetle-Bot — Miniature Combat Robot [Photo](#)

Aug 2021 – Dec 2021

- Collaborated in a four-member team to build a miniature combat robot from scratch; achieved 3rd place among sophomore Mechanical Engineering students.

SOLIDWORKS Project — Functional Scooter Model

Jan 2021 – May 2021

- Completed SOLIDWORKS coursework; collaborated in a four-member team to design and 3D-print a functional scooter model.

AI Mini-Project — Basketball Outcome Prediction

Aug 2020 – Dec 2020

- Explored ML fundamentals in Python; collaborated in a three-member team to build a predictive model for basketball game outcomes.

- Led ideation and business model development in a four-member team, designing a wearable assistive device to aid visually impaired individuals.

Technical Skills

SLAM & 3D Reconstruction

- 3D Gaussian Splatting, Visuo-Tactile SLAM (GaussianFeels), Neural Point Cloud SLAM (PoP-SLAM)
- Point Clouds, Structure from Motion (SfM), RGB-D Reconstruction, Object-Centric Gaussian Mapping
- Contact-aware Gaussian population management, occlusion-aware multimodal supervision, Levenberg-Marquardt SE(3) optimisation, gsplat
- SLAM benchmarking (TUM-RGBD, Replica datasets)
- Point-cloud & Gaussian-splat registration: SE(3)/Sim(3), Gaussian-SDF residual, ICP, FPFH+RANSAC, GeoTransformer (3DMatch / 3DLoMatch); splatreg

Deep Learning & AI

- PyTorch, TensorFlow, NVIDIA CUDA (custom kernel development)
- Multimodal Learning, Differentiable Rendering
- CNN, RNN, CNN-GRU, CNN-RNN, Vision Transformer (ViT)
- ML for signal processing and predictive analytics
- Reproducible ML workflows, experiment pipeline design, model orchestration
- Ollama (local LLM inference), Genesis (robotics simulation)

Generative 3D & Diffusion Acceleration

- Training-free sampler acceleration: **HiCache++** (DMD / exponential velocity caching); HiCache (Hermite feature caching), TaylorSeer
- Image-to-3D & 3D generative models: Hunyuan3D-2 / 2.1, TRELIS v1 / v2, Meta SAM 3D, Fast-SAM3D, DiT
- 3D Gaussian Splatting registration & merging (splatreg), gsplat
- Spectral forecasting: Dynamic Mode Decomposition (DMD / Prony / Matrix-Pencil)

AI Agents, Retrieval & Formal Methods

- MCP (Model Context Protocol) tool & server development for AI agents (mathlas)
- Dense & hybrid retrieval: transformer embeddings (Qwen3), BM25, reciprocal-rank fusion
- Formal & numeric verification: Lean 4 kernel, PSLQ, OEIS, high-precision (mpmath) identification
- Uncertainty quantification & certified planning: non-exchangeable conformal prediction, incremental search (D* Lite), certificate-directed sensing (CERT-FLOW)
- Program synthesis / evolutionary search (FunSearch), sandboxed evaluation

Robotics & Middleware

- ROS, OpenCV, Open3D, COCO API
- Sensor Fusion, Real-time Inference, Path Planning, Autonomous Navigation
- Simulation & Sim-to-Real: NVIDIA Omniverse → UR5e + Allegro Hand
- Robot hardware/software integration (perception ↔ actuation APIs)

Hardware & Sensors

- DIGIT Visuo-Tactile Sensor, RGB/RGB-D Cameras, LiDAR (camera fusion)
- VectorNav IMU, Emlid RTK GPS, NVIDIA GPUs
- UR5e Robotic Arm, Allegro Hand, Raspberry Pi / Teensy / Arduino
- 3D Motion Capture (Vicon/OptiTrack)
- 3D Modelling & Printing, Machine Shop Tools

Programming Languages

- Python, C/C++ , CUDA, MATLAB, Arduino, $\LaTeX$, HTML, Django, MySQL, VHDL, Maple

Software & Tools

- Simulation & CAD: NVIDIA Omniverse, SOLIDWORKS, Blender, KiCAD
- Dev Tools: Git/GitHub, Jupyter Notebooks, Anaconda, Linux (Ubuntu), PowerShell
- Web: React.js, Flask
- Office & Misc: Microsoft Office, Matplotlib, Sklearn

Languages

- English (Fluent), Hindi (Fluent), Korean (Beginner)

Certifications, Licenses & Awards

NVIDIA Computer Vision Nanodegree

Winter 2024

Udacity in partnership with NVIDIA

- Modules: Introduction to Computer Vision, Advanced CV & Deep Learning, Object Tracking & Localisation, Cloud Computing, Neural Network Training, C++ Programming, Applied Deep Learning Projects.

Innovative Robot Technologies and their Applications [Certificate](#)

Aug 2024

K-MOOC $\times$ Seoul National University Credential ID: c65ee02c32ee88873985a799fab29ef8

GRE General Test [Report](#)

Sept 2023

Credential ID: 82736046 Blockchain ID: 809252

- Verbal Reasoning: 153 (56th percentile)
- Quantitative Reasoning: 166 (80th percentile)
- Analytical Writing: 4.0 (56th percentile)

Capstone Project — 1st Place, Most Innovative Solution [Award](#)

May 2024

Villanova University Capstone Showcase (Industry-Sponsored)

ICE Competition — 3rd Place

Dec 2020

Innovation, Creativity & Entrepreneurship Competition, College of Engineering

Beetle-Bot Competition — 3rd Place

Dec 2021

Villanova University Mechatronics Course

Dean's List [Proof](#)

Fall 2020, Spring 2021

Villanova University

UPenn Robotics Specialisation

Summer 2021

Coursera in partnership with University of Pennsylvania

- Aerial Robotics, Computational Motion Planning, Mobility, Perception, Estimation and Learning, Capstone.

Relevant Coursework

Seoul National University

- Deep Learning
- Topics in Control and Automation
- Smart Materials and Design
- Precision Metrology and Vision Inspection
- Control Systems 1
- Engineering Research Ethics and Writing Skills
- TA: Mechanical Product Design

Villanova University

- Computer Programming for Mechanical Engineering
- Mechatronics & Microcontrollers and Applications
- Dynamic Systems I & Lab
- Capstone Design I & II
- ME Undergraduate Research I
- Manufacturing Engineering
- Mechanics of Materials / Solid Mechanics & Design I
- Electrical Circuit Fundamentals
- Material Science I
- Thermodynamics
- Engineering Interdisciplinary Project I
- Intro to Computer-Aided Design & Craft (SOLIDWORKS)
- Statics; M.E. Analysis & Design; ME Laboratory II
- Fundamentals of Computer Programming Engineering I & Lab

Yonsei University (Exchange)

- Heat Transfer, Fluid Mechanics, Mechanical Vibrations, Mechanical Systems Control, Basic Circuit Theory, Probability and Random Variables

Leadership & Activities

Secretary

Villanova CubeSat Club

- Fostered community engagement in space science with a focus on CubeSats.

May 2021 – May 2023

Villanova University

Representative & Head of Special Events

Villanova International Students' Organisation (VISO)  [Certificate](#)

- Built connections between upperclassmen and incoming international first-year students; organised social events to strengthen community.

Aug 2020 – Aug 2022

Villanova University

Member

Society of Asian Scientists and Engineers (SASE)

- Supported SASE's mission of preparing Asian-heritage scientists and engineers for the global business landscape.

May 2021 – Aug 2022

Villanova University

Member

American Society of Mechanical Engineers (ASME)

- Contributed to professional development activities and chapter engineering initiatives.

May 2021 – Aug 2022

Villanova University